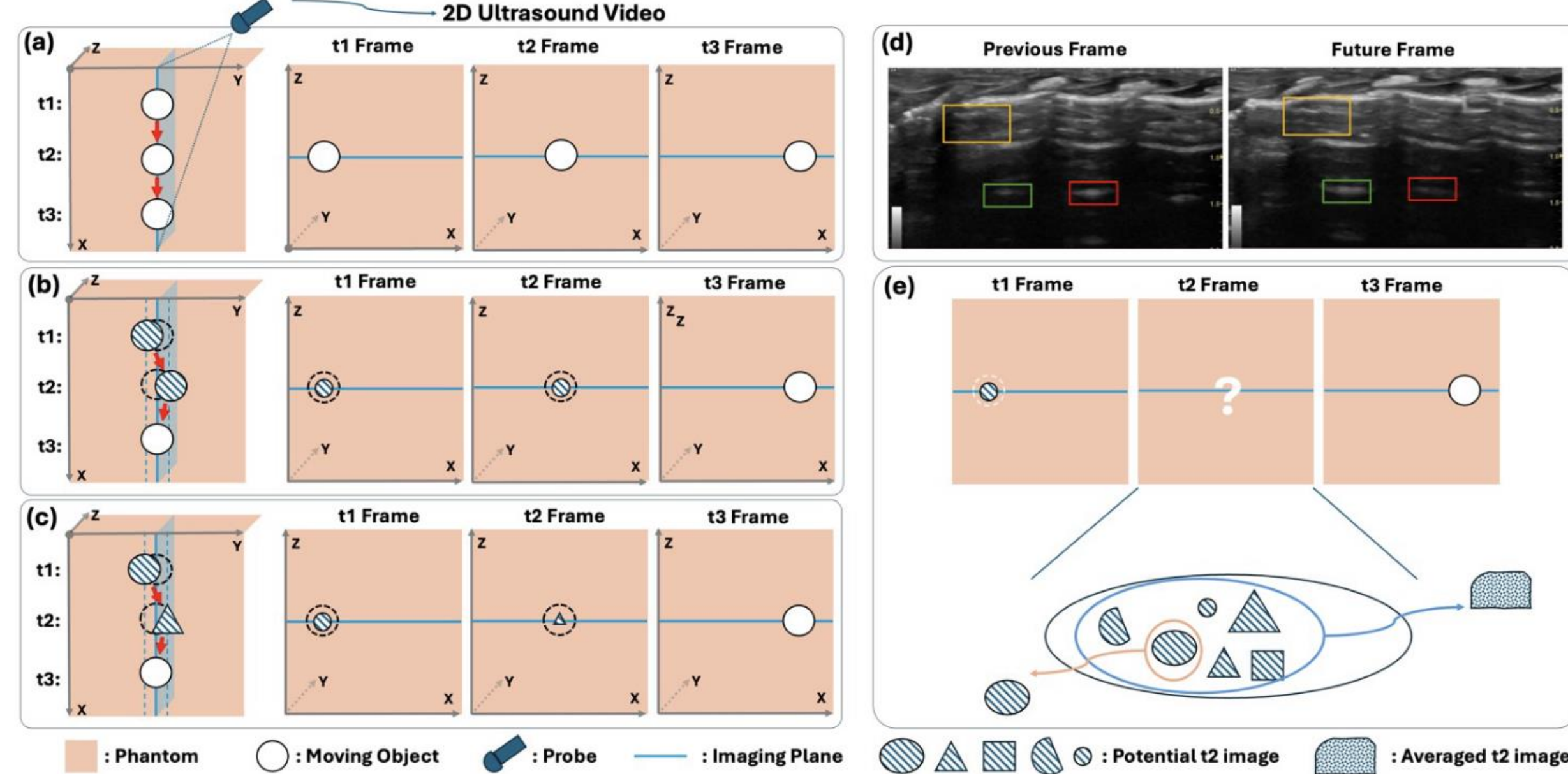


Highlights

- **Background:**
 - High temporal resolution in ultrasound (US) videos is essential for imaging fast-moving structures and accurately detecting diseases. Learning-based US Video frame interpolation (VFI) methods have not been fully investigated.
- **Challenges:**
 - Motion estimation-based methods may not be well-suited for US due to US video imaging process.
- **What we do:**
 - We formulate the US VFI problem as a generative task and introduce conditional latent diffusion model (LDM) on US VFI task.
 - Proposed method noticeably improves the perceptual metrics when evaluated in one open-source dataset and one GEHC data.

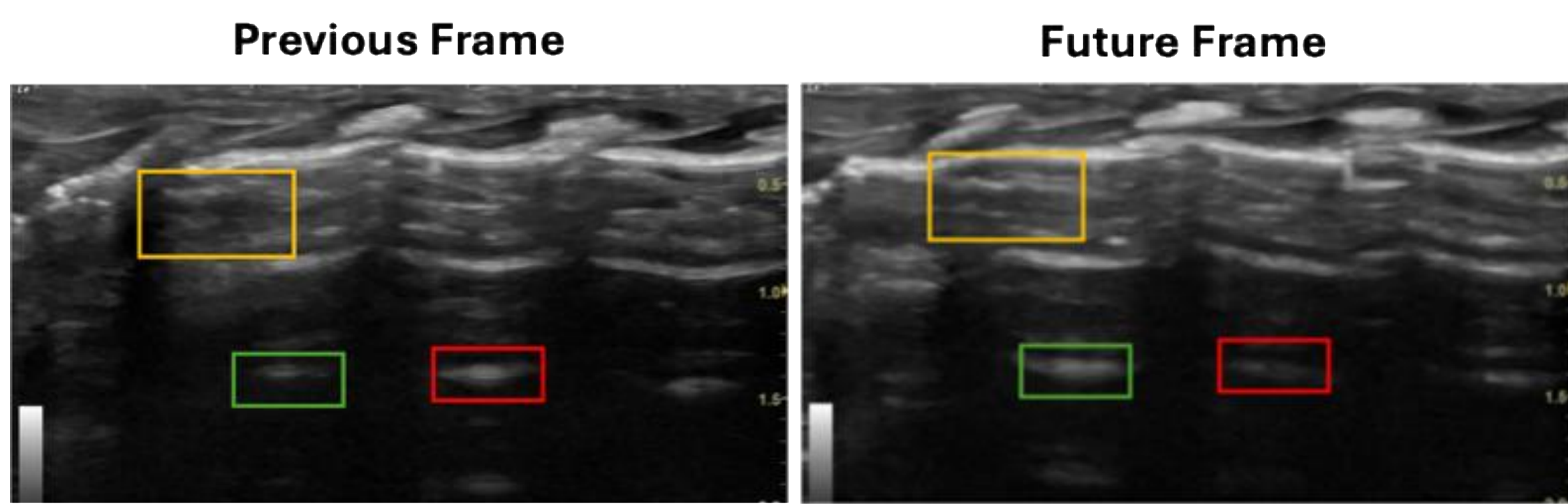
Motivation

Unique Challenge for VFI on US videos



- Each ultrasound frame represents a section between a 2D plane wave (blue shadow) and 3D tissue (phantom).
- Irregular movements cause motion perpendicular to the plane wave, leads to the change of object size in adjacent 2D video frames.
- Organ deformation leads to morphological changes in adjacent frames.
- Complex motion patterns in adjacent US video frames.
- These complex motions lead to multiple potential intermediate frames, and challenge motion-estimation-based VFI methods that focus on estimating the accurate motion of each pixel, resulting in blurry predictions (blue circle).

Challenge of Motion Estimation-Based US VFI



Motion Estimation-Based VFI methods have two assumption: 1) Gray-level invariance assumption and 2) Neighborhood optical flow similarity assumption. However, the complex motion patterns in US video frames challenge both assumptions:

- Yellow Block:** tissue deformation between US video frames.
- Red Block:** tissue in the previous frame disappears in the future frame.
- Green Block:** future frame contains unseen structure in previous frame.

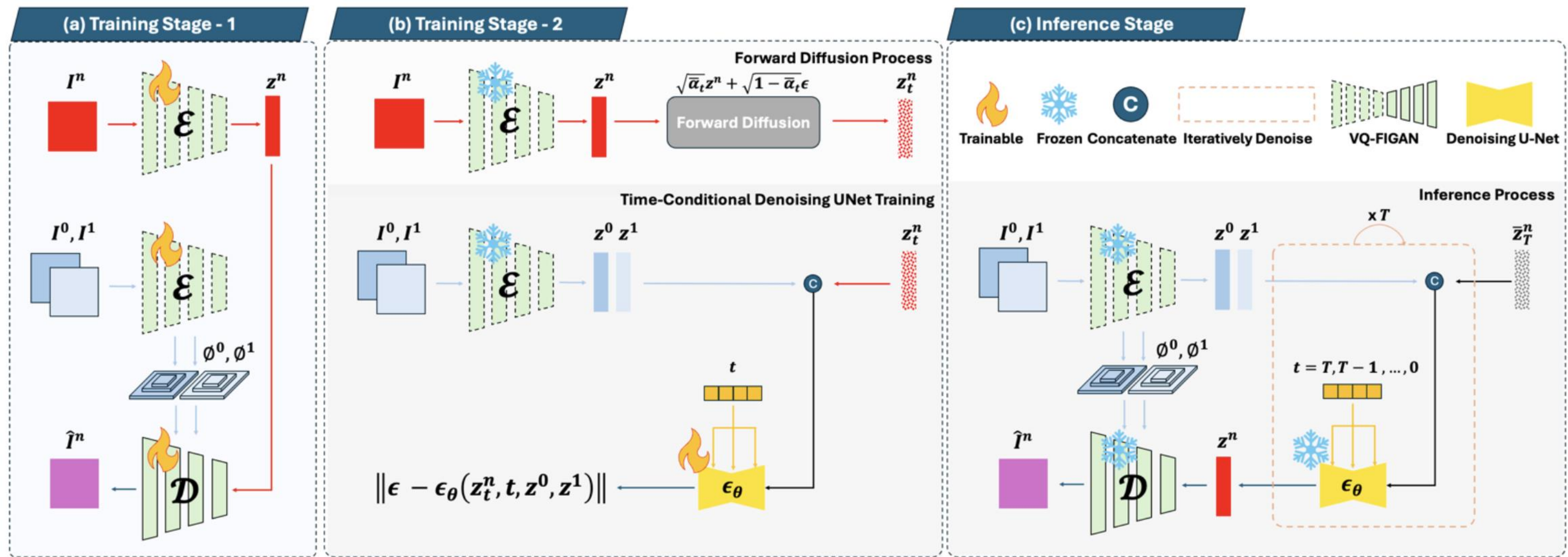
Gray-level invariance assumption ❌

Neighborhood optical flow similarity assumption ❌

Reference

- [1] Danier, Duolikun, Fan Zhang, and David Bull. "Ldmvfi: Video frame interpolation with latent diffusion models." *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 38. No. 2. 2024.
- [2] Huang, Zhewei, et al. "Real-time intermediate flow estimation for video frame interpolation." *European Conference on Computer Vision*. Cham: Springer Nature Switzerland, 2022.
- [3] Kong, Lingtong, et al. "Dynamic Frame Interpolation in Wavelet Domain." *IEEE Transactions on Image Processing* (2023).

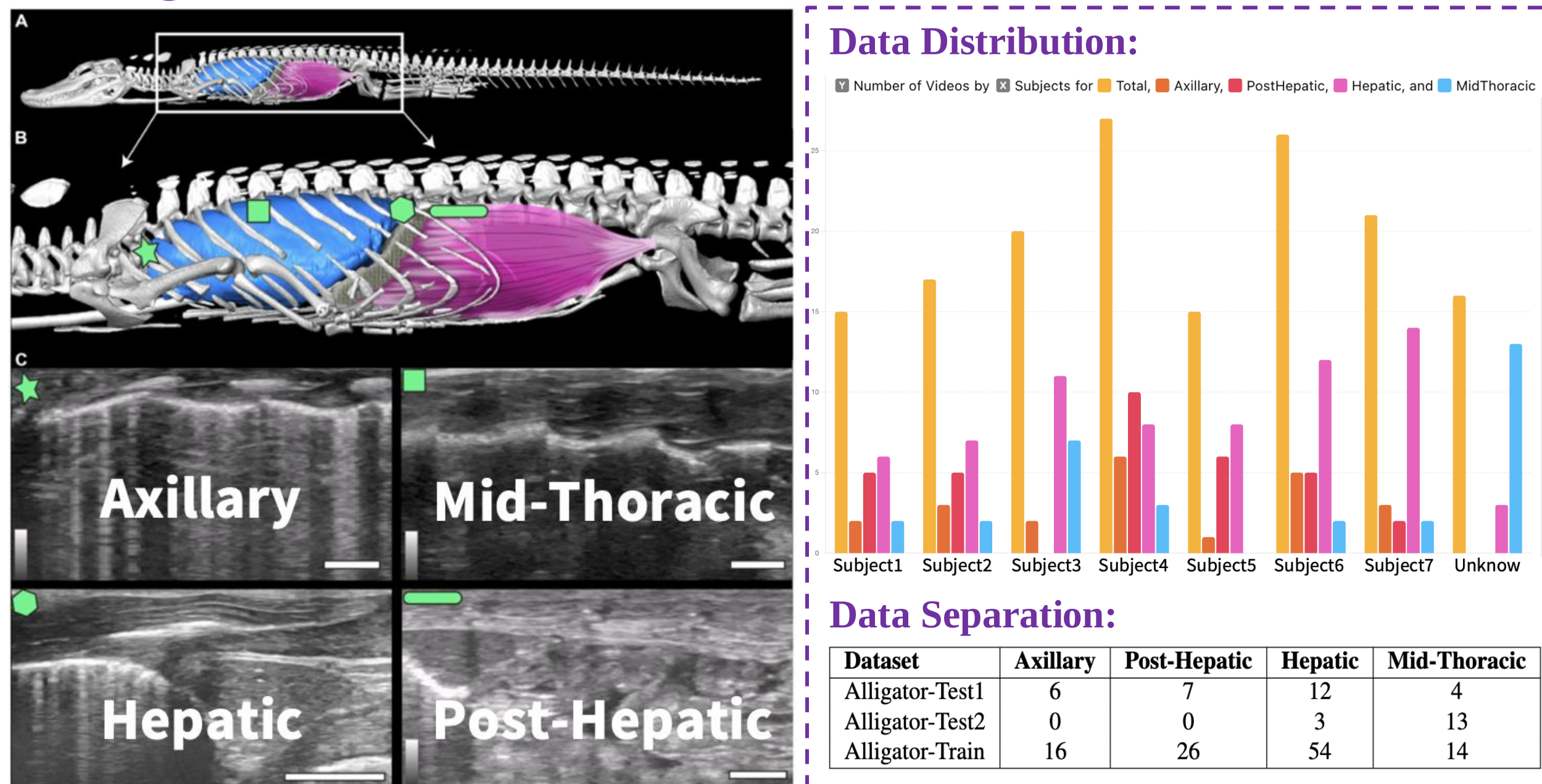
Latent Diffusion Model for Video Frame Interpolation (LDMVFI)



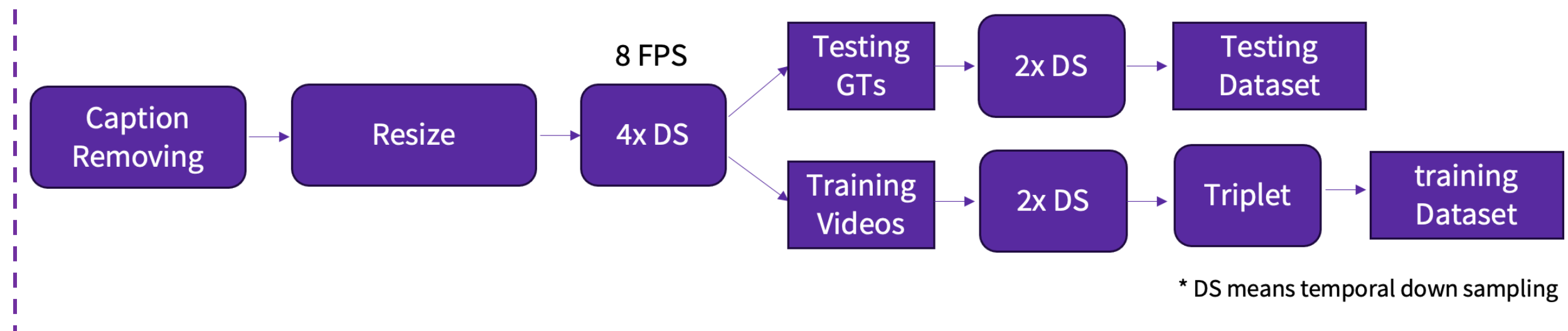
- **Training Stage-1:** we train the autoencoder for conditional image reconstruction. All the latent features are combined at multiple layers of the decoder using Cross Attention blocks. Finally, the reconstructed intermediate frame is generated conditionally.
- **Training Stage-2:** we focus on two main steps: forward diffusion and conditional denoising. During the denoising process, we combine the noisy latent feature and the conditioning latents as input for the denoising U-Net.
- **Inference Stage:** we conditionally generate the intermediate frame between each pair of adjacent frames to achieve $2 \times$ interpolation.

Dataset and Preprocessing

Alligator Dataset:



Preprocessing Pipeline:

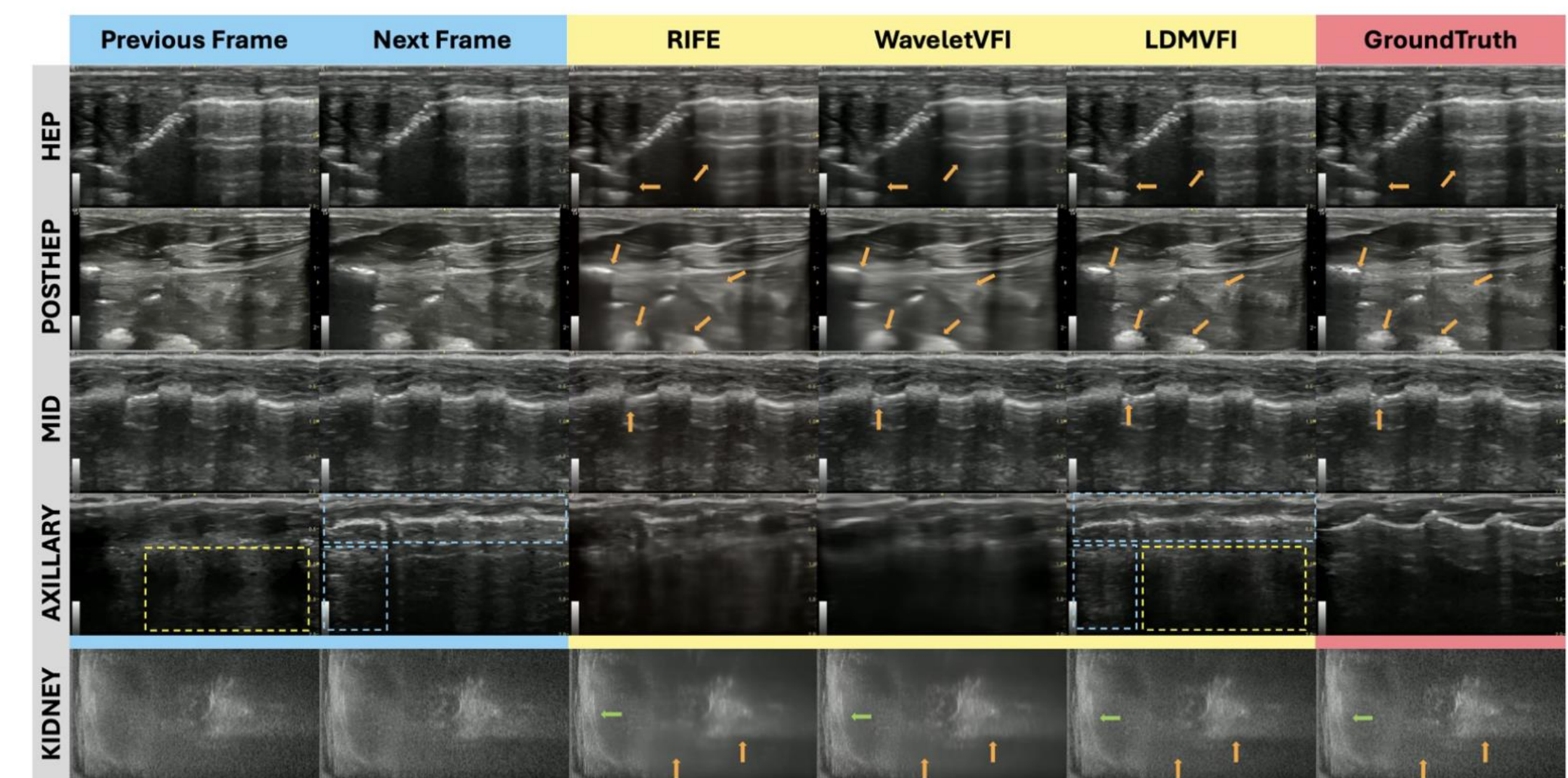


Dataset	Cropping	Frame Size	FPS	Frame Range
All Videos	Central crop 0.75 height	256 × 448 × 3	-	-
Alligator-Test1 Groundtruth	-	-	8 (downscaled 4×)	88 to 350
Alligator-Test2 Groundtruth	-	-	8 (downscaled 4×)	122 to 223
Alligator-Test1 VFI Testing	-	-	4 (downscaled 8×)	44 to 175
Alligator-Test2 VFI Testing	-	-	4 (downscaled 8×)	61 to 111
Alligator-Train	-	-	4 (downscaled 8×)	-
Training Videos Divided into non-overlapping triplet datasets $\mathcal{T} = \{X_m^0, X_m^1, X_m^2\}_{m=1}^M$				
Total Triplet Groups (M)	2698			

GEHC-Kidney Dataset:

Description	Details
Dataset Name	GEHC-Kidney
Collected By	GE Healthcare (GEHC)
Video Clips	36
Division	Evenly divided into training and testing sets
Frame Size	448 × 256 × 3
Testing Videos	
- Original Frame Rate	16 FPS
- Downsampled Frame Rate	8 FPS
- Frames per Video	15 to 24 frames
Training Videos	
- Downsampled Frame Rate	8 FPS
- Triplet Groups	117 non-overlapping

Results



- Orange and green arrows highlight regions with noticeable differences in the results of different algorithms.
- In all the samples, the baselines results contain blurry areas, while proposed method reconstructs clear details and closely resembling the ground truth.

VFI Methods	Alligator-Test1					
	FID↓	LPIPS↓	FVD↓	FLoLPIPS↓	PSNR↑	SSIM↑
RIFE	29.09 ± 11.71	0.11 ± 0.09	347.76 ± 213.86	0.08 ± 0.03	32.99 ± 3.95	0.86 ± 0.06
WaveletVFI	30.15 ± 13.34	0.11 ± 0.09	353.72 ± 304.66	0.07 ± 0.03	33.41 ± 4.08	0.90 ± 0.07
LDMVFI	14.80 ± 6.30	0.06 ± 0.05	92.74 ± 56.24	0.05 ± 0.02	33.21 ± 4.55	0.87 ± 0.08
VFI Methods	Alligator-Test2					
	FID↓	LPIPS↓	FVD↓	FLoLPIPS↓	PSNR↑	SSIM↑
RIFE	25.12 ± 7.04	0.09 ± 0.04	375.75 ± 286.92	0.06 ± 0.01	33.20 ± 1.96	0.91 ± 0.03
WaveletVFI	23.46 ± 6.87	0.08 ± 0.04	294.66 ± 247.73	0.05 ± 0.01	33.52 ± 1.93	0.91 ± 0.03
LDMVFI	13.37 ± 1.87	0.05 ± 0.01	93.09 ± 40.89	0.04 ± 0.01	33.30 ± 2.20	0.90 ± 0.03
VFI Methods	GEHC-Kidney					
	FID↓	LPIPS↓	FVD↓	FLoLPIPS↓	PSNR↑	SSIM↑
RIFE	92.38 ± 19.63	0.50 ± 0.06	881.31 ± 191.15	0.29 ± 0.04	28.35 ± 0.91	0.59 ± 0.05
WaveletVFI	108.04 ± 24.82	0.52 ± 0.07	847.43 ± 225.29	0.28 ± 0.04	28.72 ± 0.88	0.60 ± 0.05
LDMVFI	37.76 ± 11.69	0.15 ± 0.03	210.05 ± 71.02	0.09 ± 0.01	27.44 ± 1.01	0.54 ± 0.07

- Dramatic improvement on perceptual metrics indicating the proposed method results are more realistic and closer to human judgment.
- PSNR and SSIM tend to favor blurry outputs and cannot fully reflect reconstruction performance.